SCAN · PORTFOLIO

Rakesh Kumar

PRINCIPAL SYSTEM TEST ENGINEER
NETWORK VALIDATION & AUTOMATION

CONTACT

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CAREER TIMELINE

2018 – Present
Principal System Test Engineer
Keysight Technologies · Calabasas, CA

2010 – 2018
Senior QA Engineer II
Ixia Communications · Calabasas, CA

2007 – 2010
QA Engineer I
Ixia Technology Pvt Ltd · Kolkata, India

PROTOCOL EXPERTISE TREE

- **L2 — Data Link**
  - Ethernet · VLAN 802.1q
    - STP 802.1d/w · LAG/LACP
- **L3 — Network**
  - TCP/IP · ARP · ICMP

Rakesh Kumar

PRINCIPAL SYSTEM TEST ENGINEER · NETWORK VALIDATION & AI AUTOMATION SPECIALIST · 19 YEARS @ KEYSIGHT/IXIA

ABOUT

19 years at Keysight/Ixia (May 2007 – May 2026) specialising in high-performance network validation, L4–L7 traffic generation, and Python automation. Architected **BPS-MCP / AppSec MCP** (agent control of BreakingPoint & CyPerf), **LabVault** (inventory, CapEx, DCIM), and **OCS Lab UI** (optical topology ops). Deep expertise in DDoS, SSL/TLS proxy, ADC, and security testing with Palo Alto, Fortinet, and F5 BIG-IP Viprion.

- **Results-driven** — logical and broad in scope; delivers under time-crunch schedules
- **Builder** — full-stack Python, Django, FastAPI deployed to production
- **Customer-facing** — WebEx sessions with enterprise clients; translates field issues into engineering fixes
- **Self-motivated** — willingly assumes responsibilities and leads features end-to-end
- **Team-oriented** — strong co-ordination across QA, Dev, and PM with high sense of ownership
- **Mentor** — feature lead role; guides junior engineers through test design and debugging
- **Convergence belief** — integrates networking, security, AI, and automation into unified frameworks
- **Clear communicator** — thorough bug descriptions with high-level debug context; oral and written

TECHNICAL SKILLS

KEYSIGHT PLATFORMS

- IxLoad · IxNetwork · IxOS
- BreakingPoint (BPS) · BPS-QT
- Xcellon-Ultra-NP · XT80
- CloudStorm 10G/40G/100G

PROGRAMMING

- Python · TCL · Bash
- FastAPI · Pydantic · SQLAlchemy
- Django · REST APIs · MCP
- PyPDF2 · Pandas · Regex
- ChromaDB · Qdrant · RAG

NETWORK HARDWARE

- Cisco 6500 L3 · 7200 Router
- Arista L2/L3/L4–L7
- F5 BIG-IP Viprion (B4100/4200/4300)
- Palo Alto 7060 · Fortinet 3700D/2600F
- Cisco ASA · Shunra Impairment

OSPF · BGP · NAT/PAT
└NAT64 / NAT46

◆ **L4–L7 — Application**

HTTP/1.1 · HTTP/2 · HTTP/3
QUIC · TLS 1.3 · SSL Proxy
SIP/H.323 · VoIP · FTP
└DHCP · DNS · Multicast

◆ **Security**

DDoS · IPS/IDS · ATI
802.1x · Firewall Policy
└SSL Inspection · DLP

◆ **WAN / Wireless**

LTE 3GPP · GTPv2
└PPP · L2TP · MIPv4/v6

CERTIFICATIONS

- Network Management in Linux (XLRI)
- Cryptography (IIT Kharagpur)
- Cloud Computing (Texas Univ)
- Kubernetes & Containers
- AWS / GCP / Azure

- APS M8400 / M1010
- ThreatArmor · Perfect Storm

- LLM Fine-tuning · QLoRA

- Juniper · SONiC / Dell ONIE

DEVOPS & CLOUD

- Docker · Kubernetes · Ansible
- Jenkins · GitLab CI/CD
- AWS · GCP · Azure
- nginx · systemd · PXE/TFTP
- Perforce · Bitbucket · Git

SECURITY & MONITORING

- DDoS · IPS/IDS · ATI
- SSL/TLS Proxy · DLP
- 802.1x · RADIUS (Cisco ACS)
- Wireshark · NetMon
- Exploit/Malware replay (ATI)

TRAFFIC & WAN

- L4–L7 Load Balancing / ADC
- NetSecOpen standard algorithms
- LTE 3GPP · GTPv2 · PMIP
- PPP · L2TP · MIPv4/MIPv6
- P2P · Application Replay

PROFESSIONAL EXPERIENCE

Principal System Test Engineer (QA Engineer-5) — Feb 2018 – Present · Calabasas, CA

Keysight Technologies — Ixia Solutions Group

- Lead validation for high-speed load modules (Xcellon-Ultra-NP, XT80, CloudStorm 10G/40G/100G/400G) for data-centre and carrier-grade networks
- Architected BPS-MCP / AppSec MCP — agent gateways for BreakingPoint and CyPerf (55+ tools, RAG, Cursor skills, operator console)
- Built LabVault: chassis inventory, CapEx workflow, DCIM/Hyperview polling, reservations, and BPS deployment jobs across 50+ multi-vendor devices
- Delivered OCS Lab UI / ocsctl for Calient optical circuit-switch topology plan/commit and lab fiber operations
- Developed Pythar V2: Python regression framework orchestrating 154 BPT scenarios across HTTP/3, QUIC, TLS 1.3, DDoS, and application-layer protocols
- Collaborate with Product Management on customer requirements; technical input influences test platform roadmaps and feature prioritisation
- Work with Docker, Kubernetes, AWS/GCP/Azure for test-environment deployment; mentor junior engineers on automation and debugging

Senior QA Engineer II — L4–L7 Application & Security Testing

July 2010 – Jan 2018 · Calabasas, CA

Ixia Communications (HQ)

- Architected test scenarios for L4–L7 traffic management, ADC load-balancing, and application delivery (F5 BIG-IP Viprion, Cisco ACE)
- Reduced regression cycle time 40% with Python/TCL automation frameworks; spearheaded performance calibration for hyperscale data-centre customers
- Validated SSL proxy, NAT/PAT/NAT64/NAT46, enterprise firewalls (Palo Alto, Fortinet, Cisco ASA); implemented NetSecOpen standard algorithms (published internal IP)
- Conducted IPS/IDS validation with Advanced Threat Intelligence (ATI) on BreakingPoint Security platform

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QA Engineer I — Network Testing & Automation

May 2007 – July 2010 · Kolkata, India

Ixia Technology Pvt Ltd

- Built foundational expertise in L2–L7 protocol testing using Ixia IxLoad and IxNetwork
- Automated OS deployment and configuration management via TCL scripting in PXE environments
- Validated switching, routing, and VLAN implementations on Cisco, Arista, and Juniper platforms

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PUBLISHED PROJECTS

BPS-MCP — Autonomous Test Controller

PRODUCTION

MCP server exposing 55+ tools enabling AI agents (Cursor/Claude) to control BreakingPoint chassis via natural language. Features 7 Cursor skills, ChromaDB RAG knowledge base, and autonomous test-orchestration capabilities.

`55+ tools` `7 Cursor skills` `ChromaDB RAG` `15 k+ LOC`

View project →

LabVault — Network Lab Asset Management

PRODUCTION

Django system of record for 50+ multi-vendor chassis (Arista, FortiGate, PaloAlto, SONiC): IxOS/KCOS health, reservations, Hyperview-style DCIM, CapEx request→finance→procurement→asset linking, and BPS deployment jobs.

50+ devices | CapEx workflow | DCIM | Django + PostgreSQL

Live demo →

AppSec MCP — BPS + CyPerf Operator Gateway

PRODUCTION

Unified MCP gateway and web console so AI agents operate BreakingPoint and CyPerf application-security test stacks from one surface — hardware sessions, RAG knowledge, chatter telemetry.

BPS + CyPerf | MCP gateway | Console UI

OCS Lab UI — Optical Circuit Switch Tooling

PRODUCTION

Calient OCS lab UI and ocsctl: topology plan/commit, fiber inventory, AresONE mappings, guided optical cross-connect operations for multi-chassis labs.

OCS | Topology compiler | FastAPI + UI

BPS Operator — Live Regression Portal

PRODUCTION

Django ASGI web portal for Pythar V2 regression execution — browse BPT catalog, kick off test runs, stream live logs, and download reports. Integrated with LabVault for resource reservation.

154 BPTs | Daphne ASGI | Live log streaming

Live portal →

Pythar V2 — BPS Regression Framework

ACTIVE

Python framework orchestrating 154 BPT scenarios across HTTP/3, QUIC, TLS 1.3, DDoS, and application-layer protocols. Provides BPT catalog management, test-execution orchestration, and HTML performance reporting.

154 BPTs | 200+ tests | HTML reports

AI-Patents Super Engine — RAG + LLM Research Platform

ACTIVE

Full-stack AI patent research platform. RAG chatbot on fine-tuned Llama 3.2 3B (QLoRA/Unsloth), Qdrant vector database with hybrid dense + BM25 search, prior-art detection, UMAP clustering. FastAPI backend + Next.js frontend.

Llama 3.2 3B QLoRA | Qdrant | BM25 + Dense hybrid

Cogni — Persistent Autonomous Task Agent

ACTIVE

Autonomous task agent with encrypted distributed storage (Fernet AES-128 over file/HTTP/IPFS), knowledge base with confidence scoring, learning-from-experience, and pattern extraction. Demonstrates secure AI agent architecture.

`Fernet AES-128` `IPFS backend` `Confidence scoring`

EDUCATION

Post Graduate Diploma — Cloud Computing
Texas University · Great Learning

B.E. Computer Science — 73% Aggregate
BIT Sindri, Dhanbad (Four-Year Full-Time) · Jharkhand, India